THE CARTIER EQUALITY FOR A FINITELY GENERATED FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- A finitely generated field extension is the fraction field of a global complete intersection. Algebra, Lemma \ref{algebra-lemma-colimit-syntomic}, cited at proof lines 8580-8581. Provides the presentation used in the computation.
- Homotopy invariance of the naive cotangent complex and its behaviour under localisation. Algebra, Lemma \ref{algebra-lemma-NL-homotopy} and Algebra, Lemma \ref{algebra-lemma-localize-NL}, cited at proof lines 8590-8591. Identify the naive cotangent complex of the field extension with the two-term complex of the presentation.
- The dimension of a complete intersection is the number of variables minus the number of relations, and it equals the transcendence degree. Algebra, Lemma \ref{algebra-lemma-dimension-prime-polynomial-ring}, cited at proof lines 8593-8595. Gives the numerical value of the transcendence degree.

Source reference. Original lemma 07E1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9431–9431. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07BV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9355–9355. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00S1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1023–1023. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00S7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1029–1029. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00P0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 914–914. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2 (Cartier equality)**.** *Let K/k be a finitely generated field extension. Then $\Omega_{K/k}$ and $H_1(L_{K/k})$ are finite dimensional and $trdeg_k(K) = \dim_K \Omega_{K/k} - \dim_K H_1(L_{K/k})$.*

Proof. We can find a global complete intersection $A = k[x_1, \ldots, x_n]/(f_1, \ldots, f_c)$ over k such that K is isomorphic to the fraction field of A, see Algebra, Lemma 07BV and its proof. In this case we see that $NL_{K/k}$ is homotopy equivalent to the complex

$$\bigoplus\nolimits_{j=1,\ldots,c} K \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} K\mathrm{d}x_i$$

by Algebra, Lemmas 00S1 and 00S7. The transcendence degree of K over k is the dimension of A (by Algebra, Lemma 00P0) which is $n - c$ and we win. □